Apar Pokhrel

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Work Experience

Research Assistant **May 2025 – Present**
PRIMaL Lab | Python, PyTorch, CUDA, NumPy, SciPy, MMGeneration *Arlington, Texas*

- Engineered GPU-accelerated neural reconstruction pipelines for 3-D game assets using diffusion-guided Neural Point Clouds, improving reconstruction fidelity, geometric consistency, and training convergence across complex scenes.
- Develop neuromorphic vision pipelines for event-camera perception, integrating self-supervised learning and spatiotemporal representation learning for robust low-latency scene understanding in autonomous driving settings.
- Design physics-informed heat kernel diffusion frameworks for event-stream representation learning, targeting segmentation, classification, and depth estimation tasks on neuromorphic datasets including Gen1 and 1MPX.

AI Corps Fellow **January 2026 – May 2026**
GitLab | Ruby on Rails, Vue, PostgreSQL, Redis, Elasticsearch *Remote*

- Implement enhancements to the Kubernetes Agent Server (KAS) API to expose cluster infrastructure metadata, including API endpoints, node IP ranges, and network CIDR, improving infrastructure visibility for K8 deployments.
- Resolve stale Elasticsearch migration state issues by redesigning cache invalidation and lookup logic, enabling improved status accuracy with real-time propagation of $<$ 2s update latency across GitLab Admin Search UI and CLI tooling.
- Contribute to GitLab billing workflows by implementing backend and UI support for GitLab Duo's AI usage tracking and credit consumption, ensuring accurate monitoring for usage and cost attribution for 1M+ usage events/day.

Robotics Research **August 2025 – December 2025**
Robotic Vision Laboratory | Python, PyTorch, ROS/ROS2, OpenCV *Arlington, Texas*

- Integrated and fine-tuned Vision-Language-Action model (SmolVLA) policies in robotic manipulation tasks using LeRobot framework for pick-and-place tasks, targeting real-time control and policy deployment on SO-101 Arm.
- Leveraged LeRobot community datasets to benchmark and optimize SmolVLA policies, developing custom metrics for task success, trajectory efficiency, and goal accuracy to maximize manipulation performance.

Software Engineer **September 2022 – February 2024**
SAE International | Java, Spring, JavaScript, Oracle, PostgreSQL, AWS *Warrendale, Pennsylvania*

- Optimized WMI/WMC code allocation and cancellation workflows for 1000+ car manufacturers across 80+ countries, achieving 70% faster lookup and assignment by implementing indexed PostgreSQL tables and caching strategies.
- Engineered a serverless PDF reporting pipeline on AWS Lambda with SendGrid API and S3, automating report generation, transactional email delivery, and real-time notifications, cutting distribution time by 60%.
- Integrated Keycloak for user auth and SSO with role-based access for 10,000+ users, implementing OIDC/OAuth2 token flows and automated provisioning, and documented API endpoints in Swagger UI for seamless integration.

Technical Skills

Languages: Python, Java, TypeScript, SQL, C/C++
Robotics & Vision: ROS/ROS2, OpenCV, CUDA, TensorRT, ONNX, LeRobot, IsaacLab
MLOps:: PyTorch, SLURM, CUDA, DDP, Weights & Biases, FiftyOne, Roboflow
Database & APIs: Oracle, MySQL, PostgreSQL, MongoDB, OpenAPI/Swagger, FastAPI, REST
Libraries & Frameworks : ReactJS, Axios, NodeJS, ExpressJS, JAX, Hugging Face , Accelerate
Cloud & Dev Tools: AWS (EC2, S3, Lambda), GCP (Cloud Storage, Firebase), Git, GitLab CI/CD, Docker, ELTK

Projects

Humanoid Skill-Blending RL | *Python, IssacLab* **March 2026 – Present**

- Develope a humanoid skill-blending framework for whole-body loco-manipulation in simulation using composable primitive skills and extend to vision-based RL with ego-centric RGB observations using PPO and DreamerV3 baselines.

Parking Occupancy Detection – PKLot | *Python, PyTorch*

- **[Paper]** Evaluated YOLOv8 backbones (ResNet-18, VGG16, EfficientNet, Ghost-P2), achieving up to 98.6% mAP50:95, 0.9–4.1ms inference latency, 47% reduction in model parameters with computational efficiency as low as 8.2 GFLOPs.

Education

The University of Texas at Arlington **May 2026**
Master of Science in Computer Science, Machine Learning & Database Systems *GPA: 4.0/4.0*

The University of Texas at Arlington
Bachelor of Science in Computer Science, Minor in Mathematics *GPA: 3.81/4.0*